UC Berkeley
Teaching Professor
Dan Garcia

CS61C

Great Ideas in Computer Architecture
(a.k.a. Machine Structures)

Memory (Mis)Management

cs61c.org

Agenda

Endianness, Word Alignment

Memory and Addresses

How to read byte addresses:
`0x...FFFE` `0x...FFFC`

- Modern machines are "byte-addressable."
  - Hardware's memory composed of 8-bit storage cells; each byte has a unique address
- We commonly think in terms of "word size":
  - aka number of bits in an address
  - A **32b** architecture has **4-byte words**.
  - All pointer sizes on 32b architecture:

```
sizeof(int *) ==  ...
     == sizeof(char *) == 4
```

	+3	+2	+1	+0
0xFFFFFFFC	int32_t *			
0xFFFFFFF8	short *			
0xFFFFFFF4	char *			
0xFFFFFFF0	xx	xx	xx	xx
0xFFFFFFEC	xx	xx	xx	xx
0xFFFFFFE8	xx	xx	xx	xx
...	xx	xx	xx	xx
0x0C	xx	xx	xx	xx
0x08	xx	xx	xx	xx
0x04	xx	xx	xx	xx
0x00	xx	xx	xx	xx

Endianness

- The hive machines are "little endian".
  - The **least significant byte** of a value is stored first.
- (Contrast with "big endian")
  - (Most significant byte is stored first)

`int32_t*` pointer
0xFFFFFF**F0**

`int32_t` value
0x123456**78**

	+3	+2	+1	+0
0xFFFFFFFC	0xFF	0xFF	0xFF	0xF0
0xFFFFFFF8	short *			
0xFFFFFFF4	char *			
0xFFFFFFF0	0x12	0x34	0x56	0x78
0xFFFFFFEC	xx	xx		
0xFFFFFFE8	xx	xx	xx	
...	xx	xx	xx	xx
0x0C	xx	xx	xx	xx
0x08	xx	xx	xx	xx
0x04	xx	xx	xx	xx
0x00	xx	xx	xx	xx

Word Alignment

Read for HW01

- We often want "word alignment":
  - Some processors will not allow you to address **32b** values without being on **4-byte** boundaries.
  - Others will just be very slow if you try to access "unaligned" memory.

	+3	+2	+1	+0
0xFFFFFFFC	int32_t *			
0xFFFFFFF8	short *			
0xFFFFFFF4	char *			
0xFFFFFFF0	32-bit integer stored in four bytes			
0xFFFFFFEC	xx	xx	16-bit short	
0xFFFFFFE8	xx	xx	xx	8-bit char
...	xx	xx	xx	xx
0x0C	xx	xx	xx	xx
0x08	xx	xx	xx	xx
0x04	xx	xx	xx	xx
0x00	xx	xx	xx	xx

Structures, Revisited, Again

Read for HW01

- A “struct” is really an instruction to C on how to arrange a bunch of bytes in a bucket.
  - Structs provide enough space for the data.
  - C compilers often **align** the data with **padding**.

  - For this struct, the actual layout on a 32b architecture would be as follows.
    - Note the 3 bytes of padding
    - `sizeof(struct foo) == 12`

```
struct foo {
    int32_t a;
    char b;
    struct foo *c;
}
```

+3	+2	+1	+0
4 bytes for c			
unused	unused	unused	1 byte for b
4 bytes for a			

More: StackOverflow post

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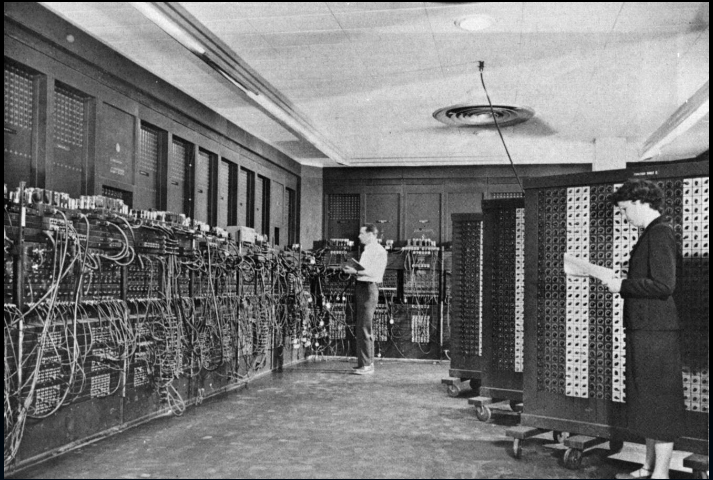
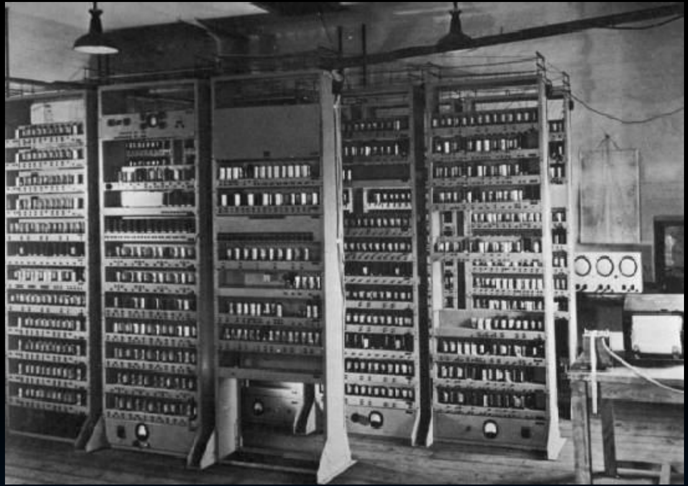
From ENIAC (1946) to EDSAC (1949)



- ENIAC: First Electronic General-Purpose Computer
- Needed 2-3 days to setup new program
- Programmed with patch cords and switches
  - At that time & before, "computer" mostly referred to people who did calculations
  - Mostly women! (See *Hidden Figures*, 2016)



- EDSAC: First General Stored-Program Computer
- Programs held as **numbers in memory**
  - Revolution! Program is also data!
- 35-bit binary **two's complement** words

What gets printed?

[Concept Check]

`sizeof()`: **compile-time operator; gives size in bytes (of type or variable).**

```
// for this exercise, assume
// shorts are 16b on a 64-bit architecture
void mystery(short arr[], int len) {
  printf("%d ", len);
  printf("%d\n", sizeof(arr));
}

int main() {
  short nums[] = {1, 2, 3, 99, 100};
  printf("%d ", sizeof(nums));
  mystery(nums, sizeof(nums)/sizeof(short));
  return 0;
}
```

A. 10 5 10
B. 10 5 8
C. 80 5 80
D. 80 5 40
E. Other

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What gets printed?

[Concept Check]

sizeof(): compile-time operator; gives size in **bytes** (of type or variable).

```
// for this exercise, assume
// shorts are 16b on a 64-bit architecture
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  short nums[] = {1, 2, 3, 99, 100};
  printf("%d ", sizeof(nums));
  mystery(nums, sizeof(nums)/sizeof(short));
  return 0;
}
```

Array has decayed to a pointer

In array's declared scope, total array size.

In array's declared scope, # elements in array.

A. 10 5 10
B. 10 5 8
C. 80 5 80
D. 80 5 40
E. Other

Agenda

Memory Locations

- Memory Locations
- The Stack
- The Heap
- Linked List Example
- When Memory Goes Bad
- Implementing Memory Management

C Program Address Space

- A program's address space contains 4 regions:
  - **Stack**: local variables inside functions, grows downward
  - **Heap**: space requested via `malloc()`; resizes dynamically, grows upward
  - **Data (Static Data)**: variables declared outside main, does not grow or shrink
  - **Text (aka code)**: program executable loaded when program starts, does not change
  - `0x00000000` chunk is unwriteable/unreadable so you that crash on `NULL` pointer access
- Programming in C requires knowing where objects are in memory, otherwise things don't work as expected.
  - By contrast, Java hides location of objects.

For now, OS somehow prevents accesses between stack and heap.
(more later w/virtual memory)

Where are variables allocated?

- **Global**: If declared **outside** a function, allocated in **data (static)** storage.
- **Local**: If declared **inside** function, allocated on the **stack** and freed when function returns.
  - NB: `main()` is also a function.
- For both these memory types, the management is **automatic**.
  - You don't need to worry about deallocating when you are no longer using them.
  - But a variable **does not exist anymore** once a function ends!

```
int myGlobal;
… main() {
   int myTemp;
…
}
```

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Agenda

The Stack

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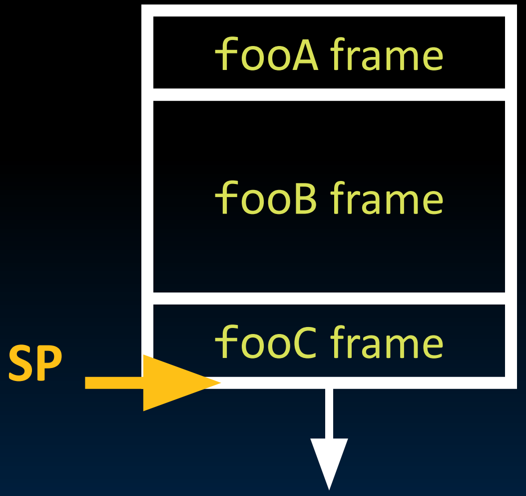
The Stack

- Every time a function is called,
  a new **stack frame** is allocated on the stack.
- A stack frame includes:
  - Return "instruction" address (who called me?)
  - Arguments*
  - Space for other local variables
- Stack frames contiguous blocks of memory;
  the **stack pointer** indicates the start of stack frames.
- When function ends, stack frame is tossed off the stack; frees memory for future stack frames.
- (more later when we cover details for a RISC-V processor architecture)

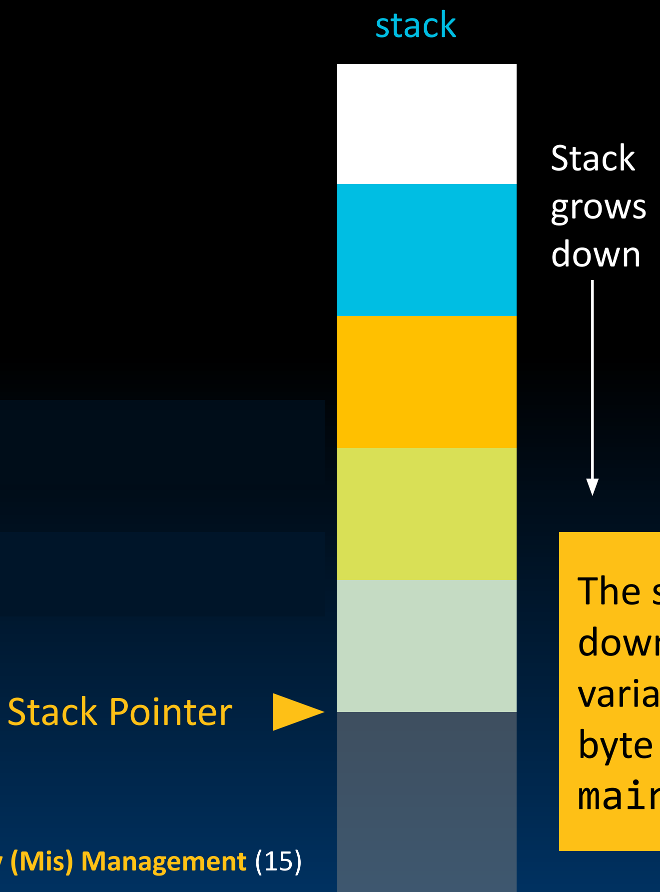
```
fooA() { fooB(); }
fooB() { fooC(); }
fooC() { … }
```



*see slide notes in pptx for technical elaboration

The Stack is Last In, First Out (LIFO)

```
int main ()
{ a(0);  …
}
  void a (int m)
  { b(1);
  }
    void b (int n)
    { c(2);
    }
      void c (int o)
      { d(3);
      }
        void d (int p)
        {
        }
```



The stack grows downward; `a()`'s local variables have lower byte addresses than `main()`'s, and so on.

Kao

Recall: Array Are Very Primitive

1. Array bounds are not checked during element access.
   - Consequence: We can accidentally access off the end of an array!
2. An array is passed to a function as a pointer.
   - Consequence: The array size is lost! Be careful with `sizeof()`!
3. Declared arrays are only allocated while the scope is valid.

```
char *foo() {
   char string[32]; ...;
   return string;
}
```
is incorrect

Solution:
Dynamic memory allocation!
(more ~~late~~ now)

Passing Pointers into the Stack

✅ It is fine to pass a pointer to stack space further down.

- I.e., pointers to addresses higher in the stack point to data in currently allocated stack frames.

```
void load_buf() { … };
int main() {
   …
   char buf[...];
   load_buf(buf, BUFLEN);
   …
}
```

stack

buf char array persistent through load_buf's execution

❌ **However, it is catastrophically bad to return a pointer to something in the stack!**

- Memory will be overwritten when other functions are called!
- So your data would no longer exist…and writes can overwrite key pointers, causing crashes!

```
char *make_buf() {
    char buf[50];
    return buf;
}
void foo() {…}
int main(){
   char *stackAddr = \
      make_buf();
   foo(stackAddr);
   …
}
```

stack

stackAddr points to overwritten memory

~~buf???~~

Agenda

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The Heap

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What is the Heap?

- The heap is dynamic memory – memory that can be allocated, resized, and freed during program runtime.
  - Useful for persistent memory across function calls.
  - But biggest source of pointer bugs, memory leaks, …
  - Similar to Java **new** command allocates memory….but with key differences below.
- **Huge** pool of mem (usually >> stack), but not allocated in contiguous order.
  - Back-to-back requests for heap memory *could* result in blocks very far apart
- In C, specify number of bytes of memory **explicitly** to allocate/deallocate item.
  - `malloc()`: Allocates raw, uninitialized memory from heap
  - `free()`: Frees memory on heap
  - `realloc()`: Resizes previously allocated heap blocks to new size
  - Unlike the stack, memory gets reused only when programmer **explicitly** cleans up

void *malloc(size_t n)

- Allocates a block of uninitialized memory:
  - `size_t n` is an unsigned integer type big enough to "count" memory bytes.
  - Returns `void *` pointer to block of memory on heap.
  - A return of NULL indicates no more memory (**always** check for it!!!)
- To allocate a struct:

```
typedef struct { ... } TreeNode;
TreeNode *tp = (TreeNode *) malloc(sizeof(TreeNode));
```

Typecast casts return value from type `(void *)` to `(TreeNode *)`

`sizeof(type)` gives size in bytes.

- To allocate an array of 20 `int`s:

```
int *ptr = (int *) malloc(20*sizeof(int));
if (ptr != NULL) { ... // always check NULL after
```

  - Many years ago `int`s used to be 16b. Now, 32b or 64b…

Assuming size of objects can lead to misleading, unportable code. Use `sizeof()`!

void free(void *ptr)

- Dynamically frees heap memory
  - `ptr` is a pointer containing an address originally returned by `malloc()`/`realloc()`.

```
int *ptr = (int *) malloc (sizeof(int)*20);
...
free(ptr); // implicit typecast to (void *)
```

- When you free memory, be sure to pass the original address returned from `malloc()`. Otherwise, crash (or worse!)

void *realloc(void *ptr, size_t size)

- Resize a previously allocated block at `ptr` to a new size.
  - Returns new address of the memory block.
    - In doing so, it may need to copy all data to a new location.
  - `realloc(NULL, size); // behaves like malloc`
  - `realloc(ptr,  0); // behaves like free, deallocates heap block`
- Remember: Always check for return NULL,
  which would mean you've run out of memory!

```
int *ip; ip = (int *) malloc(10*sizeof(int));
… … …                        /* check for NULL */
ip = (int *) realloc(ip, 20*sizeof(int));
/* contents of first 10 elements retained */
… … …                        /* check for NULL */
realloc(ip,0);          /* equivalent to free(ip); */
```


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Agenda

When Memory Goes Bad

- Memory Locations
- The Stack
- The Heap
- Linked List Example
- **When Memory Goes Bad**
- Implementing Memory Management

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Working with Memory

✅ Code, Static storage are easy:

- They never grow or shrink.

✅ Stack space is also easy:

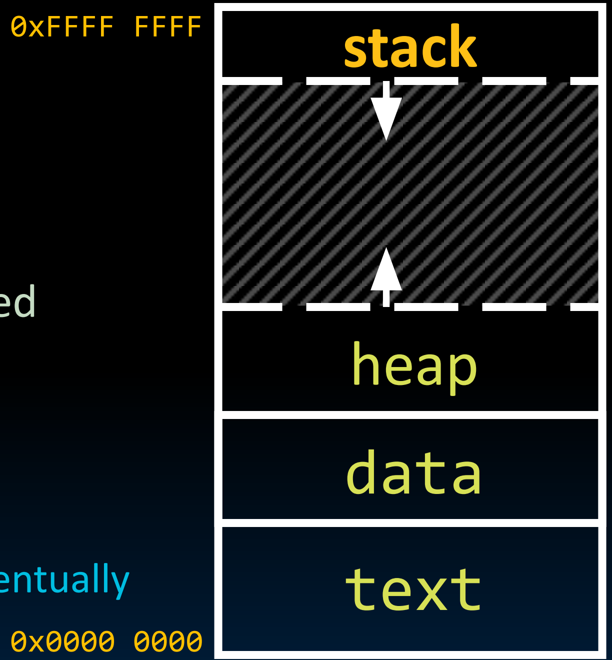
- Stack frames are created and destroyed in LIFO order.
- Just avoid "**dangling references**": pointers to deallocated variables (e.g., from old stack frames).

⚠️ Working with the heap is tricky:

- Memory can be allocated / deallocated at any time!
- **"Memory leak": If you forget to deallocate memory**
- **"Use after free": If you use data after calling free**
- **"Double free": If you call free 2x on same memory**

Your program will eventually run out of memory

Possible crash or exploitable vulnerability



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Failure to `free()`

- The runtime **does not** check for the programmer's failure to manage memory.
  - Memory is so performance-critical that there just isn't time to do this.
  - Usual result: you corrupt the memory allocator's internal structure, and you find out much later in a totally unrelated part of your code!
- **Memory leak**: Failure to `free()` allocated memory
  - 🙃 Initial symptoms. Nothing…
    - Until you hit a critical point, memory leaks aren't actually a problem
  - 🎢 …Later symptoms: performance drops off a cliff…
    - Memory hierarchy behavior tends to be great just up until it isn't, then it hits several cliffs
  - 😈 …and then your program is killed off!
    - Because the operating system (OS) says "no" when you ask for more memory

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Use after Free

- "**Dangling reference**"
  When you keep using a pointer, even after it has been deallocated
- Reads after the free may be corrupted!
  - If something else takes over that memory, your program will probably read the wrong information!
- Writes corrupt other data!
  - Uh oh… Your program crashes later!

```
struct foo *f;
…
f = malloc(sizeof(struct foo));
…
free(f);
…
bar(f->a); // !!!
```

Double-Free…

```
struct foo *f = (struct foo *) malloc(10*sizeof(struct foo));
...
free(f);
...
free(f); // !!!
```

- May cause either a use-after-`free` (because something else called `malloc()` and got that data) or corrupt heap data (because you are no longer freeing a pointer tracked by `malloc`)

Forgetting `realloc()` Can Move Data

- "Dangling reference"
- Remember, when you `realloc` it can copy data to a different part of the heap.

```
int *nums;
nums = malloc(10*sizeof(int));
...
int *g = nums;
...
nums = realloc(nums,
         20*sizeof(int));

// g could now point
// to invalid memory
```

```
int *nums;
nums = malloc(10*sizeof(int));
...
// forget to update nums
// on realloc call
realloc(nums, 20*sizeof(int));

// nums could now point
// to invalid memory,
// and we could have potentially
lost a pointer to a new block
```

  - Reads may be corrupted, and writes may corrupt other pieces of memory.

Faulty Heap Management

[Concept Check]

How many memory management errors are in this code?

[for next time]

```
void free_mem_x() {
  int fnh[3];
  ...
  free(fnh);
}

void free_mem_y() {
  int *fum = malloc(4*sizeof(int));
  free(fum+1);
  ...
  free(fum);
  ...
  free(fum);
}
```

A. 1
B. 2
C. 3
D. 0
E. Other

Faulty Heap Management

[Concept Check]

How many memory management errors are in this code?

[for next time]

```
void free_mem_x() {
  int fnh[3];
  ...
  free(fnh);
}

void free_mem_y() {
  int *fum = malloc(4*sizeof(int));
  free(fum+1);
  ...
  free(fum);
  ...
  free(fum);
}
```

free() on stack-allocated memory

free() on memory that isn't the pointer from malloc

Double free()

A. 1
B. 2
C. 3
D. 0
E. Other

Valgrind to the rescue!!!

- Valgrind slows down your program by an order of magnitude, but...
  - It adds a tons of checks designed to catch most (but not all) memory errors
    - Memory leaks
    - Misuse of free
    - Writing over the end of arrays
- Tools like Valgrind are absolutely essential for debugging C code.

And in Conclusion…

- C has 3 pools of memory for variables:
  - Data: **global/static** variable storage, basically permanent
  - Stack: **local variable** storage, parameters, return address
  - Heap (**dynamic** storage): malloc() grabs space from here, free() returns it.
  - (4th memory pool: text, for the program executable itself)
- Heap data is biggest source of bugs in C code!

Agenda

Linked List Example

- Memory Locations
- The Stack
- The Heap
- Linked List Example
- When Memory Goes Bad
- Implementing Memory Management

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Linked List Example

```c
# include <string.h>
int main() {
1   struct Node *head = NULL;
2   add_to_front(&head, "abc");
    …  // free nodes, strings here…
}
void add_to_front(struct Node **head_ptr, char *data) {
3   struct Node *node = malloc(sizeof(struct Node));
4   node->data = malloc(strlen(data) + 1); // extra byte!
5   strcpy(node->data, data);  // strcpy also copies null terminator
6   node->next = *head_ptr;
7   *head_ptr = node;
}
```

```c
struct Node {
  char *data;
  struct Node *next;
};
```

Linked List Example

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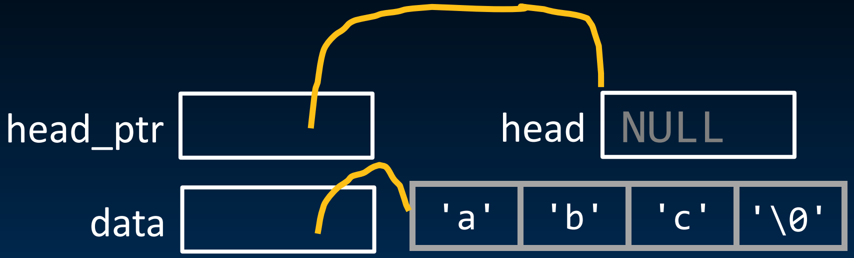
   node->next = *head_ptr;
   *head_ptr = node;
}
```

```c
struct Node {
  char *data;
  struct Node *next;
};
```



Linked List Example

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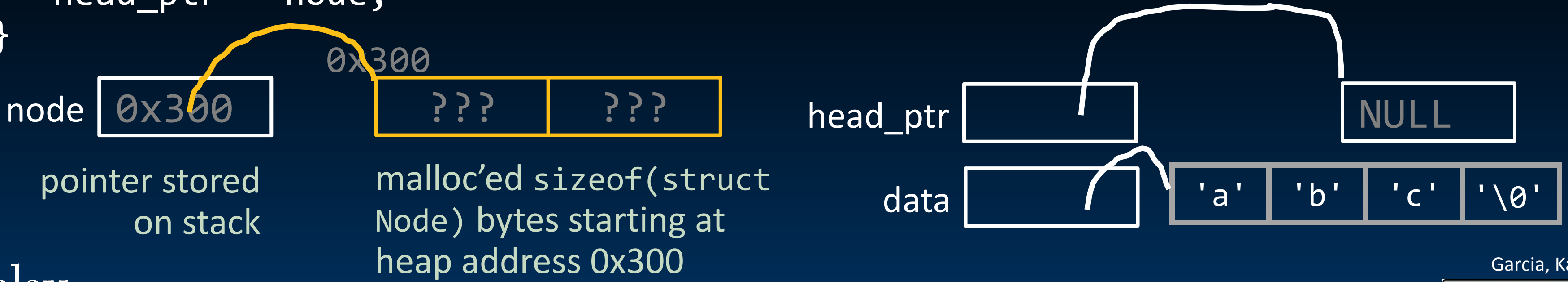
5   strcpy(node->data, data);  // strcpy also copies null terminator
6   node->next = *head_ptr;
7   *head_ptr = node;
  }
```

```
struct Node {
  char *data;
  struct Node *next;
};
```



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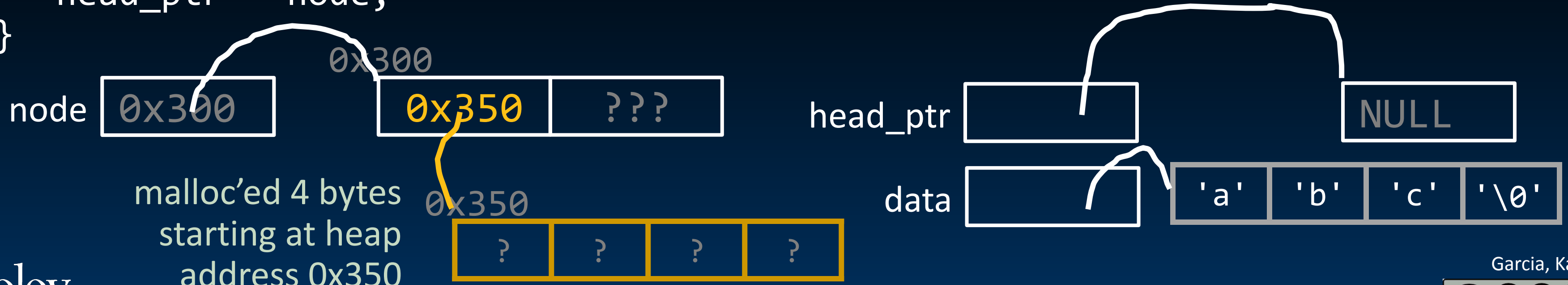
  *head_ptr = node;
}
```

```
struct Node {
  char *data;
  struct Node *next;
};
```



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Linked List Example

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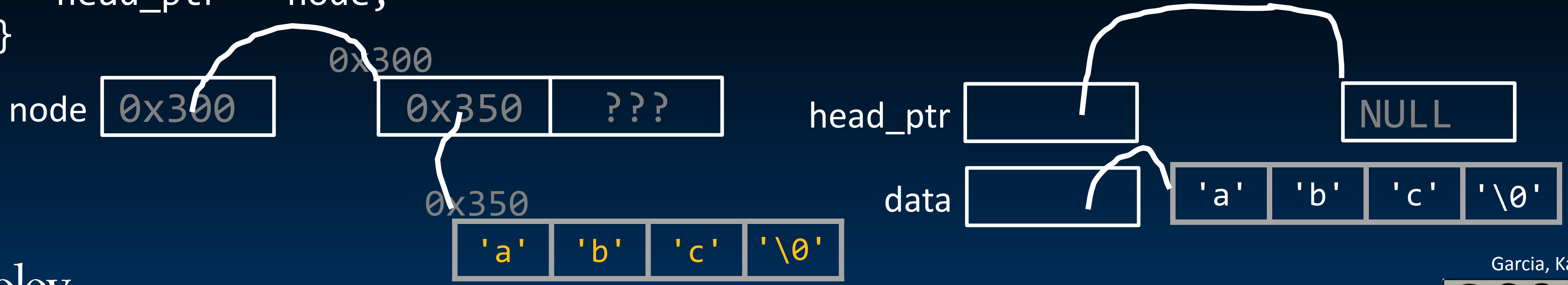
  node->next = *head_ptr;
  *head_ptr = node;
}
```

```c
struct Node {
  char *data;
  struct Node *next;
};
```



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Linked List Example

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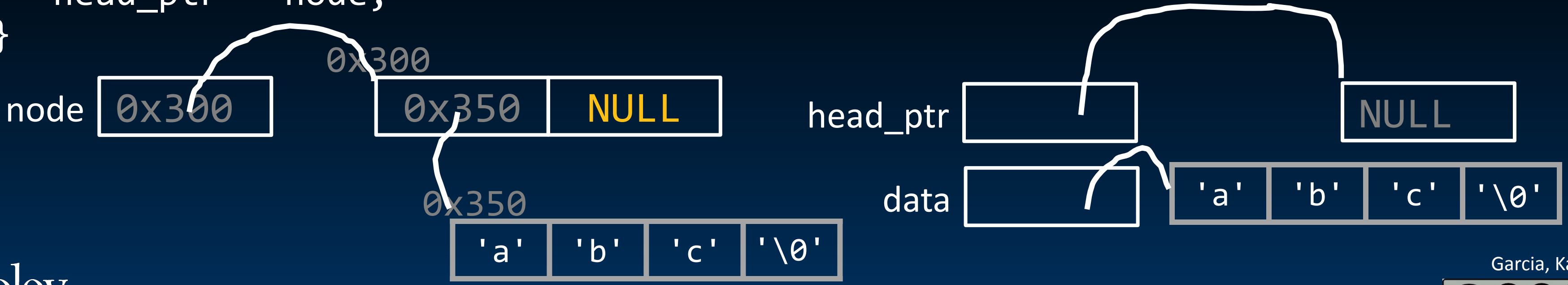
  node->next = *head_ptr;
  *head_ptr = node;
}
```

```c
struct Node {
  char *data;
  struct Node *next;
};
```



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Linked List Example

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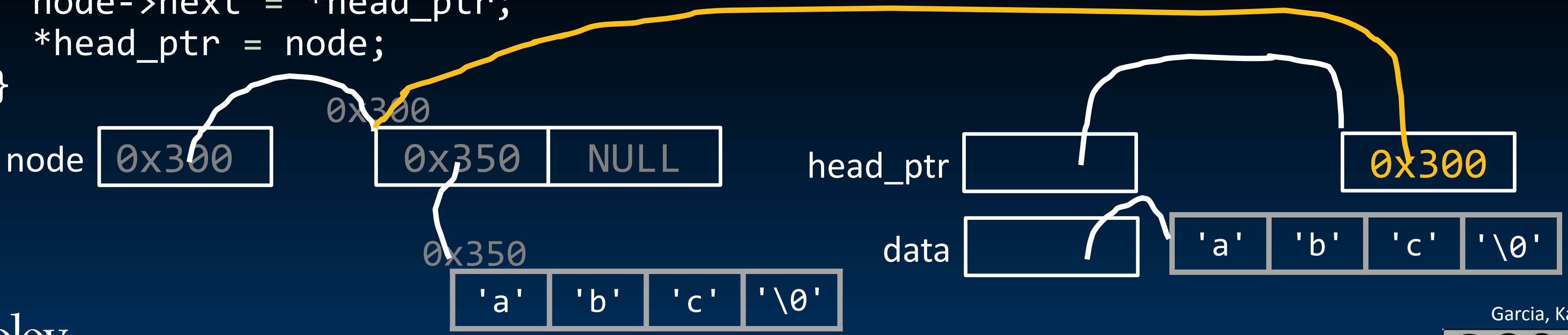
  *head_ptr = node;
}
```

```
struct Node {
  char *data;
  struct Node *next;
};
```



Linked List Example

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  node->next = *head_ptr;
  *head_ptr = node;
}
```

```c
struct Node {
  char *data;
  struct Node *next;
};
```

Check out the lecture code in Drive!

Agenda

Implementing Memory Management

- Memory Locations
- The Stack
- The Heap
- Linked List Example
- When Memory Goes Bad
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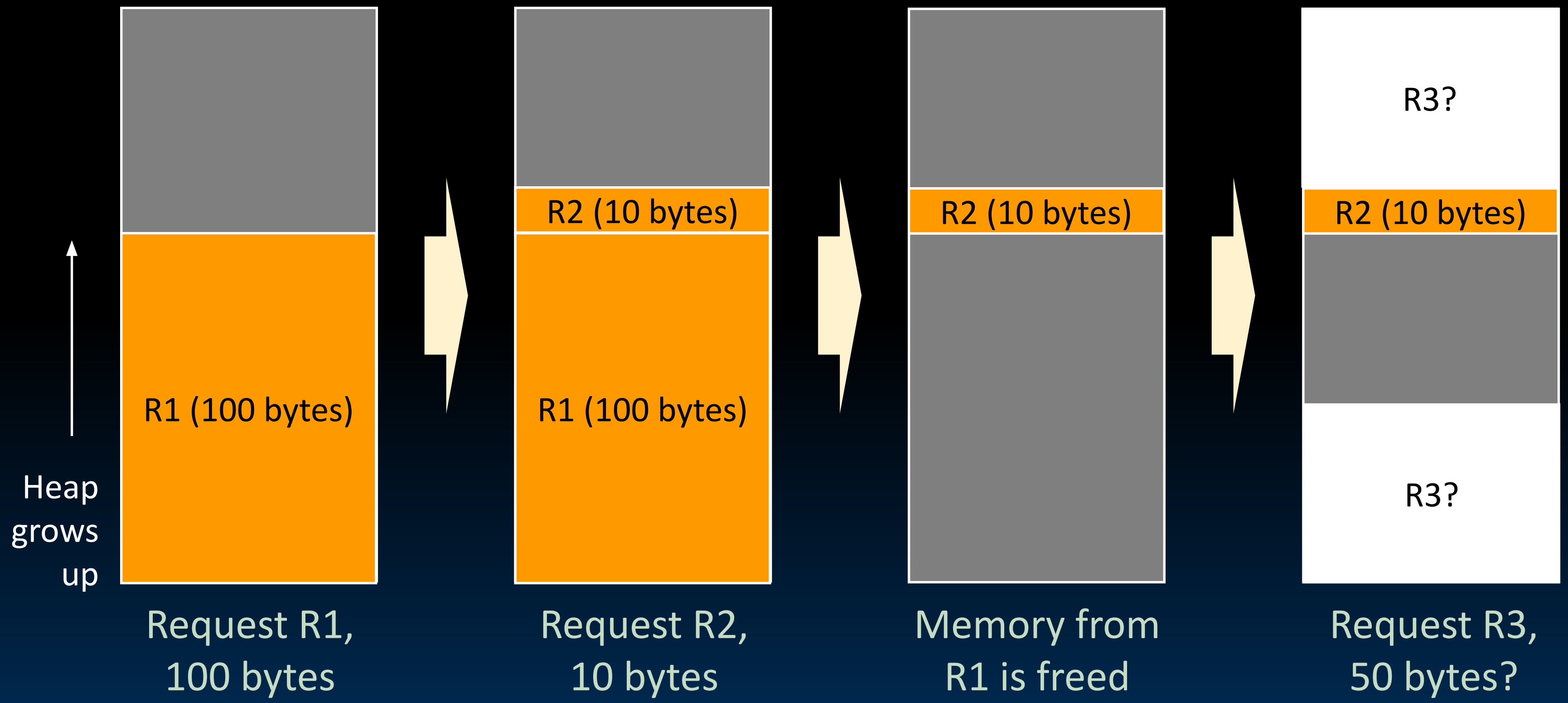
Material not tested. Recording: https://www.youtube.com/watch?v=Sq5tSeWfnGY

Heap Management Requirements

- Want `malloc()` and `free()` to run quickly
- Want minimal memory overhead
- Want to avoid fragmentation*, when most of our free memory is in many small chunks
  - In this case, we might have many free bytes but not be able to satisfy a large request since the free bytes are not contiguous in memory.

* This is technically external fragmentation

Heap Management Example



K&R Malloc/Free Implementation

- From Section 8.7 of K&R
  - Code in the book uses some C language features we haven't discussed and is written in a very terse style; don't worry if you can't decipher the code
- Each block of memory is preceded by a header that has two fields:
  - size of the block, and
  - a pointer to the next block
- All free blocks are kept in a circular linked list.
- In an allocated block, the header's pointer field is unused.

K&R Malloc/Free Implementation

- `malloc()` searches the free list for a block that is big enough. If none is found, more memory is requested from the operating system. If what it gets can't satisfy the request, it fails.
- `free()` checks if the blocks adjacent to the freed block are also free.
  - If so, adjacent free blocks are merged (**coalesced**) into a single, larger free block.
  - Otherwise, freed block is just added to the free list.

Choosing a block in `malloc()`

- If there are multiple free blocks of memory that are big enough for some request, how do we choose which one to use?
  - **best-fit**: choose the smallest block that is big enough for the request.
  - **first-fit**: choose the first block we see that is big enough.
  - **next-fit**: like first-fit, but remember where we finished searching and resume searching from there.